

Main Memory Design (SRAM)

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Main Memory

Main memory is where programs and data are kept when the processor is actively using them. When programs and data become active, they are copied from secondary memory into main memory where the processor can interact with them. A copy remains in secondary memory.

Main memory is intimately connected to the processor, so moving instructions and data into and out of the processor is very fast.

Main memory is sometimes called RAM. RAM stands for Random Access Memory. "Random" means that the memory cells can be accessed in any order. However, properly speaking, "RAM" means the type of silicon chip used to implement main memory.

There are two type of RAM. They are: Static Random Access Memory (SRAM) and Dynamic Random Access Memory (DRAM).

SRAM vs DRAM

SRAM stands for Static Random-Access Memory. They store data as do flip-flops where extra 2 transistors are used for controlling the access. It is very fast and its power consumption is also less. As long as power is being supplied to the machine, SRAM will hold data and will lose it as soon as power will be disconnected.

DRAM stands for Dynamic Random-Access Memory. It is the second type of RAM and is manufactured with the help of transistors and capacitors. The function of capacitor is to store data such that the charged capacitor indicates the value 1 and the discharged capacitor indicated a 0.

Capacitor always eventually discharges owing to the fact that charges leak. This phenomenon is shown by the word dynamic. This happens even when power is provided at all times. This is the very reason that DRAM utilizes greater amount of power. If we have to make the data last for a long period of time, there is a constant need to refresh the data. For greater amount of storage, DRAM is used. It is cheaper as only one transistor is needed for one memory block.

Main Memory (SRAM)

Question:

Suppose, you want to store 100 instructions in your main memory (RAM) and size of your ISA is 13 bits, then

1. How many address lines will be needed in RAM?
2. What should be the minimum word size of RAM so that instruction can be stored in one address?

Main Memory (SRAM)

Question:

Suppose, you want to store 100 instructions in your main memory (RAM) and size of your ISA is 13 bits, then

1. How many address lines will be needed in RAM?

Answer: No. of address lines = $\log_2 100 = 6.64 \approx 7$ bits

2. What should be the minimum word size of RAM so that instruction can be stored in one address?

Answer: Since size of ISA is 13 bits, minimum word size of RAM must be 13 bits.

Main Memory (SRAM)

RAM chip is shown below:

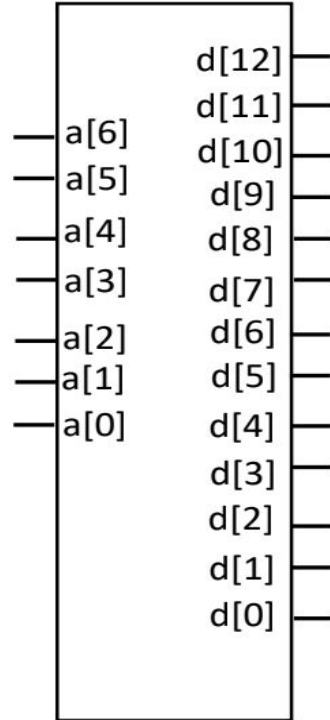


Fig. 1. RAM chip (1024 × 12)

Single port SRAM (without write operation)



Figure: RAM chip (128x13)

[illegible]

Single port SRAM (without write operation)

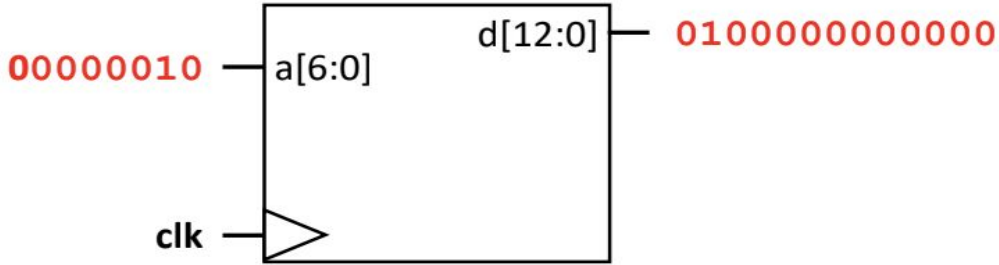


Figure: RAM chip (128x13)

[illegible]

Single port SRAM (without write operation)

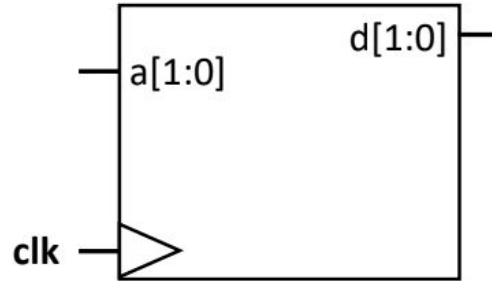


Figure: RAM chip (?x?)

Question:

1. What is the size of RAM?
2. What is the word size of RAM?

Single port SRAM (without write operation)

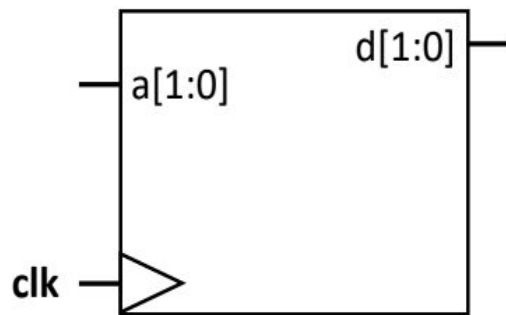


Figure: RAM chip (4x2)

Question:

1. What is the size of RAM? **Ans: 4**
2. What is the word size of RAM? **Ans: 2**

Single port SRAM (with write operation)

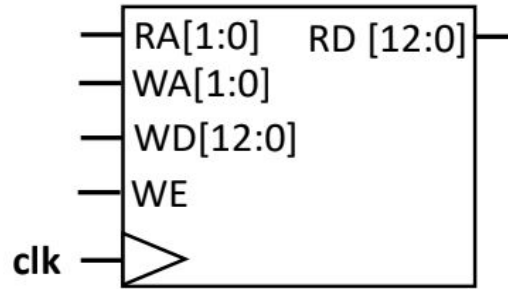


Figure: RAM chip (4x13)

Here,

RA = Read Address

RD = Read Data

WA = Write Address

WD = Write Data

WE = Write Enable

Dual port SRAM (with write operation)

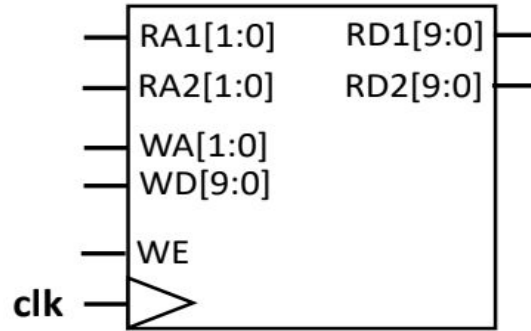


Figure: RAM chip **(4x10)**

Here,

RA = Read Address

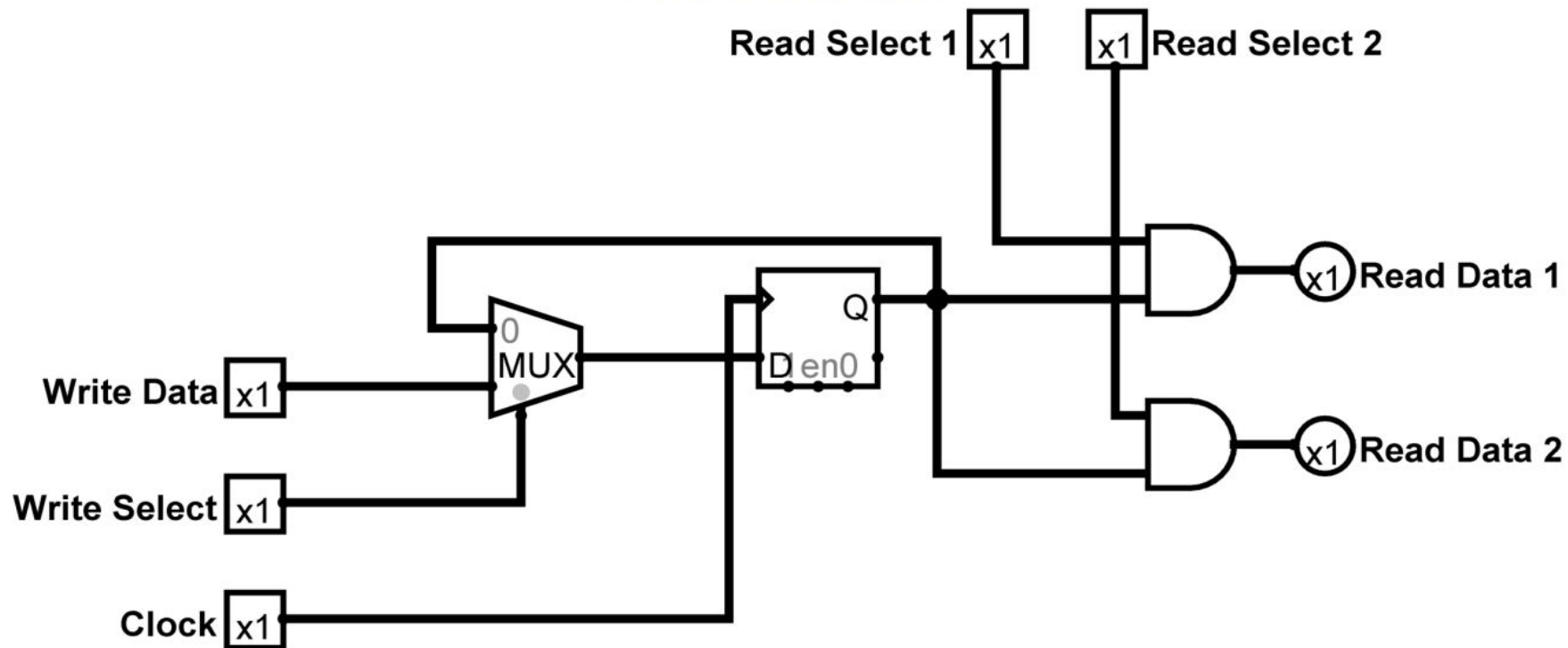
RD = Read Data

WA = Write Address

WD = Write Data

WE = Write Enable

1x1 SRAM

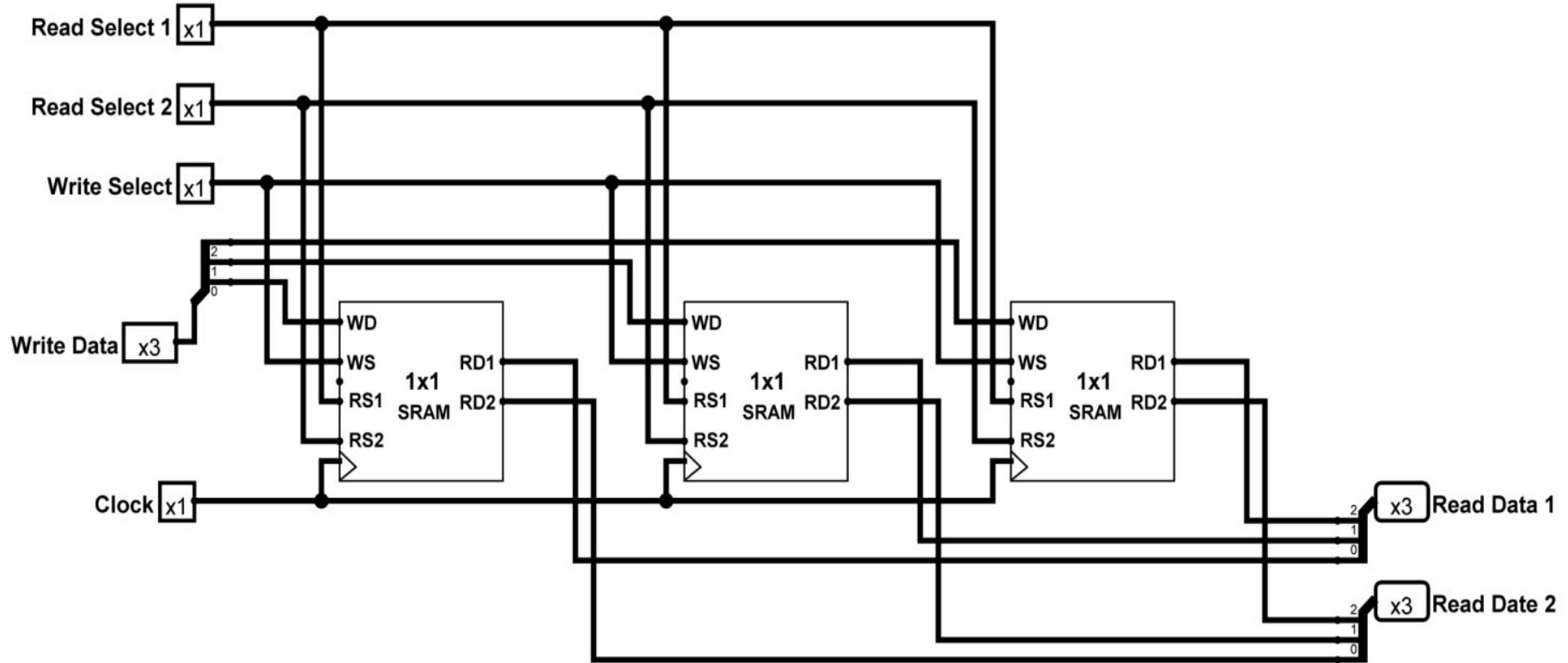


To read data from **Read data 1** of 1x1 SRAM, **Read Select 1** must be equal **1**

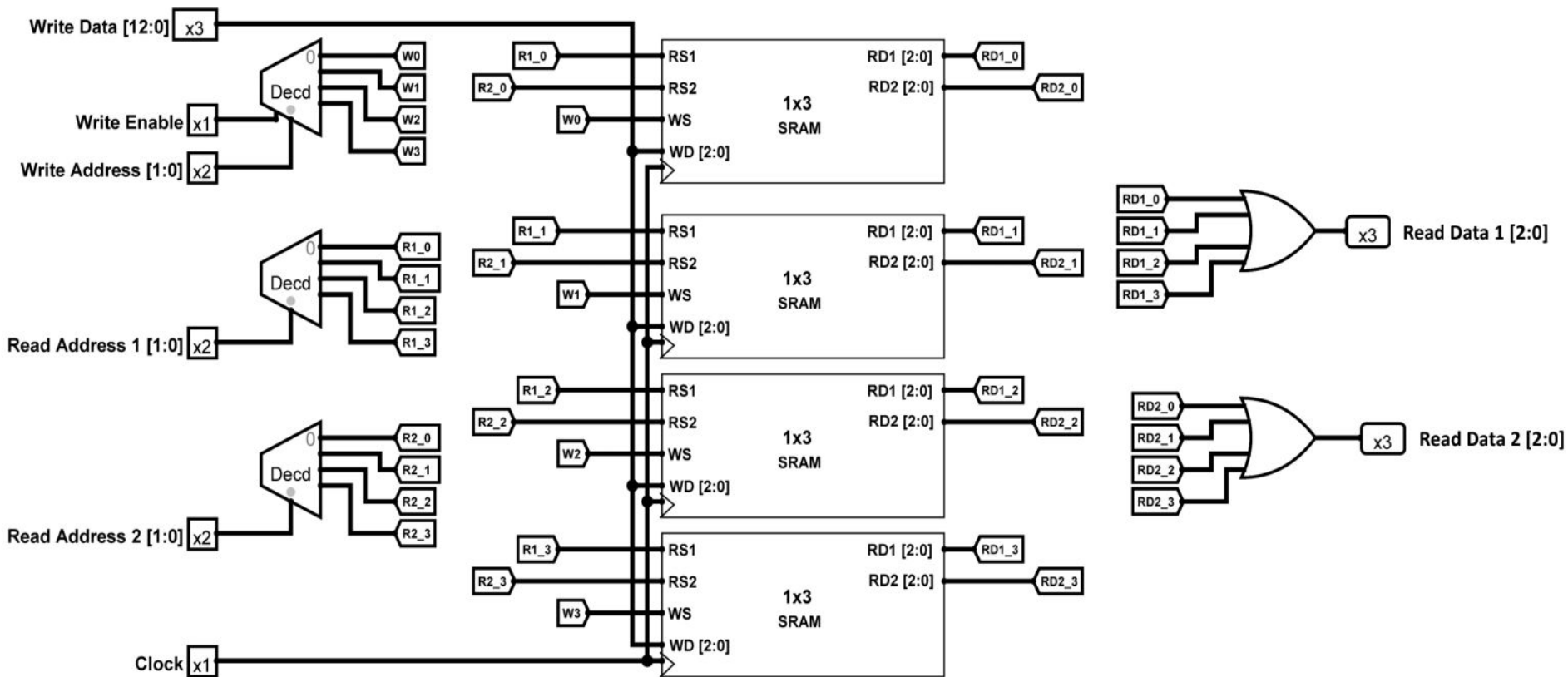
To read data from **Read data 2** of 1x1 SRAM, **Read Select 2** must be equal **1**

To write **Write data** to 1x1 SRAM, **Write Select** must be **1** and Clock must be positive edge.

1x3 SRAM



4x3 SRAM



Example: Main Memory

Question: Draw 4x3 Main Memory having Dual ports with write operation.

Answer:

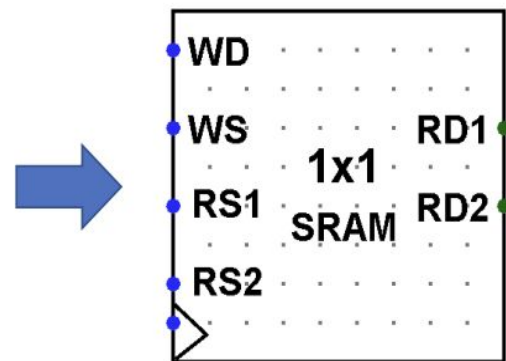
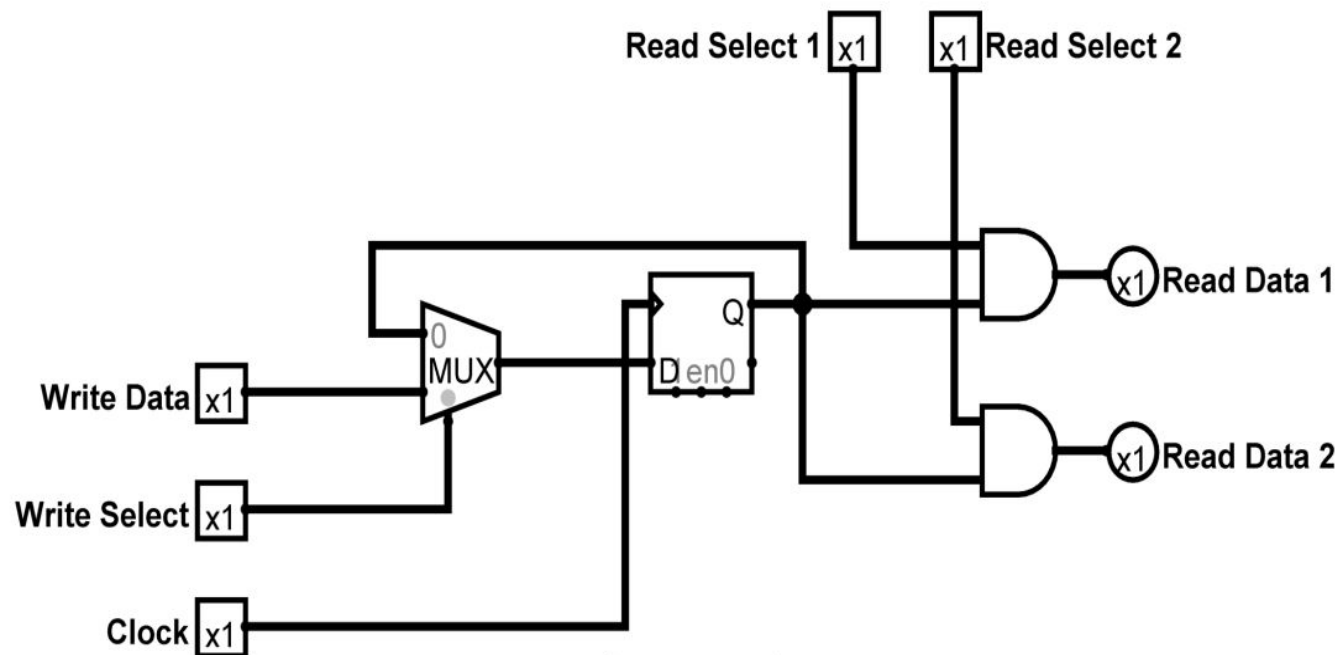


Figure: 1x1 SRAM Chip

Example: Main Memory

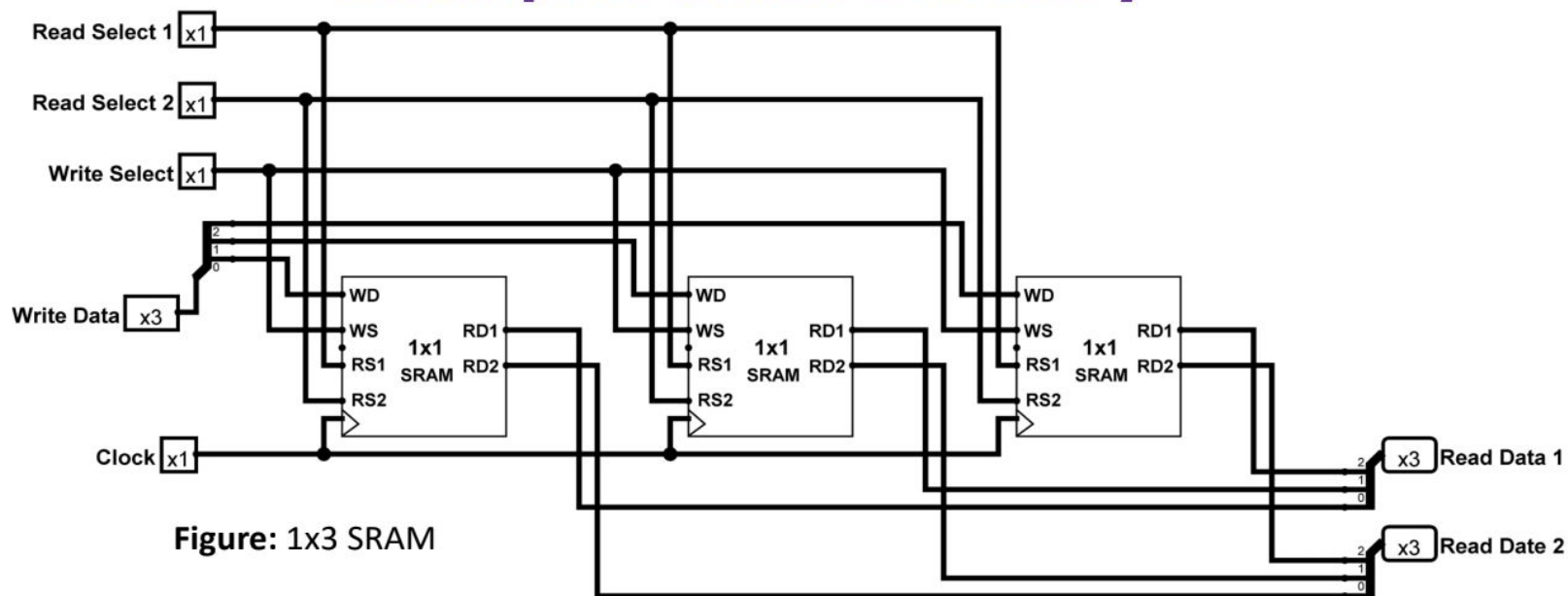


Figure: 1x3 SRAM

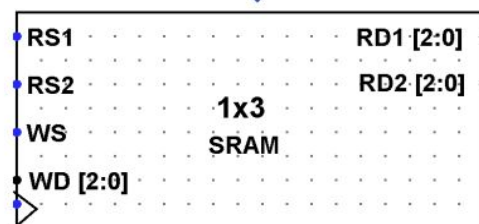


Figure: 1x3 SRAM Chip

Example: Main Memory

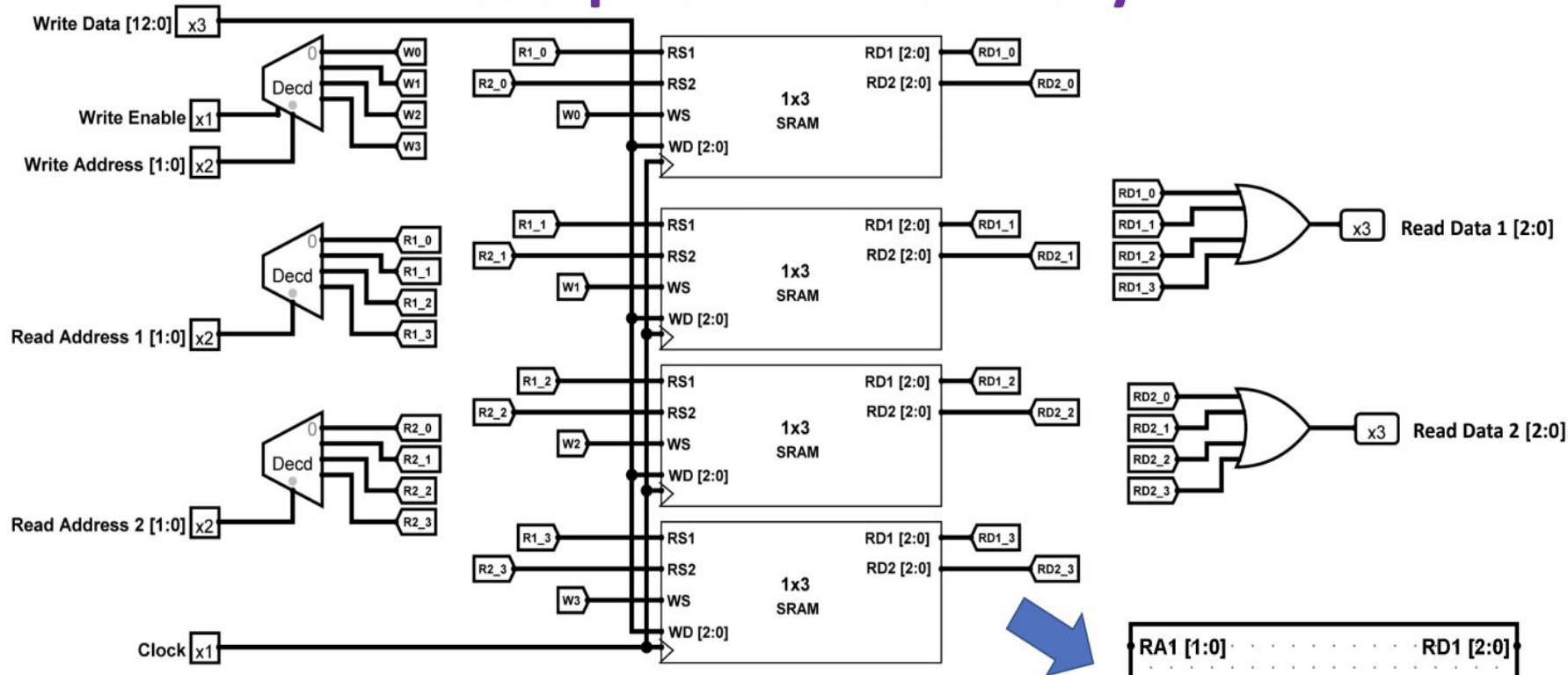


Figure: 4x3 SRAM

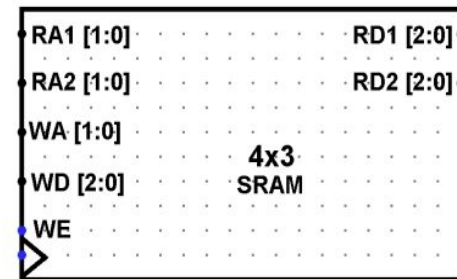


Figure: 4x3 SRAM Chip